A Project Report

ON

Electricity Bill Estimater

BY

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1. Introduction

This section will introduce the Electricity Bill Estimator project, explaining the motivation behind it, the problems it solves, and the benefits it offers to users in efficiently calculating their monthly electricity expenses.

2. Objective

The objective will outline the main goal of the project, such as providing users with an easy-to-use tool to estimate their electricity bill based on their consumption and local rates.

3. System Design and Architecture

3.1. Input Data:

This subsection will describe the input data required by the estimator, such as the number of units consumed, tariff rates, and any fixed charges.

3.2. Core Components:

The key components of the system, such as user input module, calculation engine, and result display, will be detailed here.

3.3. Data Structures:

Any data structures used to manage the input and output, like arrays or objects, will be explained.

3.4. Flow of the Program:

A step-by-step explanation of the program's workflow, from receiving user inputs to displaying the estimated bill.

4. Implementation Details

4.1. Data Structures:

Explanation of how data is stored and manipulated within the system.

4.2. Code Explanation:

Key parts of the code, highlighting how the estimation is performed and results are generated.

5. Features

A list of features the Electricity Bill Estimator offers, such as handling different tariff rates, seasonal adjustments, or additional charges.

6. Testing and Results

Details of how the system was tested and examples of results produced by the estimator.

7. Challenges and Limitations

Discussion on any challenges faced during the development and limitations of the current version of the estimator.

8. Conclusion

Summary of the project's success in meeting its objectives, along with potential future improvements or expansions.

9. CODE

```
 #include <iostream>

using namespace std;

int main() {
    double usage;
    double costPerKWh;
    double additionalFees;
    double totalBill;
    char calculateAgain;

    do {
        cout << "Welcome to the Electric Bill Calculator!" << endl;

        while (true) {
            cout << "Enter your electricity usage (in kWh): ";
            cin >> usage;
            if (usage >= 0) {
                break;
            }
            cout << "Invalid input. Please enter a non-negative number." << endl;
        }
```

```
    while (true) {
        cout << "Enter the cost per kWh: ";
        cin >> costPerKWh;
        if (costPerKWh >= 0) {
            break;
        }
        cout << "Invalid input. Please enter a non-negative number." << endl;
    }

    while (true) {
        cout << "Enter any additional fees: ";
        cin >> additionalFees;
        if (additionalFees >= 0) {
            break;
        }
        cout << "Invalid input. Please enter a non-negative number." << endl;
    }

    totalBill = (usage * costPerKWh) + additionalFees;

    cout << "\n--- Bill Summary ---" << endl;
    cout << "Electricity Usage: " << usage << " kWh" << endl;
    cout << "Cost per kWh: $" << costPerKWh << endl;
    cout << "Additional Fees: $" << additionalFees << endl;
    cout << "Total Bill: $" << totalBill << endl;

    cout << "Do you want to calculate another bill? (y/n): ";
    cin >> calculateAgain;

} while (calculateAgain == 'y' || calculateAgain == 'Y');
```

```
    cout << "Thank you for using the Electric Bill Calculator!" << endl;


    return 0;
}
```

1	**Project Title**	**ELECTRICITY BILL ESTIMATOR**		
2	**Lab**	**CYG1611- Applications of Information and Communication Technologies Lab**	**Semester**	**Fall 2024**
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4	Instructor Name & Signature	Mr. SM Waqas Ayub Shah

Project Demonstration

Assessment Criteria	Very Poor 0-1	Poor 2-3	Satisfactory 4-5	Good 6-8	Excellent 9-10	Score Student 1	Score Student 2	Score Student3
Design Evaluation	No or very poor design prototype and	Design prototype is not working	Design prototype is partially functional	Design prototype is functional and some	Design prototype is fully functional			
and Testing	demonstration.	and no testing of design has been done	and little testing of design has been done.	testing of design has been done.	and design has been exhaustively tested.			

Usage of software tools (Visual Studio, MS Office Applications) in design and evaluation	**No or very poor software tool (Visual Studio, MS Office Applications) usage in project design and results evaluation**	**Insignificant evidence of software tool (Visual Studio, MS Office Applications) usage in project design and results evaluation**	**Little evidence of ability to select appropriate software tools (Visual Studio, MS Office Applications), in project design and results evaluation**	**Some evidence of skills to use software tools (Visual Studio, MS Office Applications) in project design and results evaluation**	**Clear evidence of skills to use software tools (Visual Studio, MS Office Applications) in project design and results evaluation**			

Project Report

Assessment Criteria	Very Poor 0-1	Poor 2	Satisfactory 3	Good 4	Excellent 5	Score	Score	Score

						Student 1	Student 2	Student3
Literature Survey, Problem Analysis and Design Procedure	**No or very poor literature survey done. No problem analysis performed. No worthwhile design procedure exists.**	**Insufficient literature survey Problem analysis part is skipped or does not contribute to creating an effective design. Does not follow any design procedure.**	**Partial literature survey. Problem Analyses performed is haphazard and design parameter selection is spontaneous. Little use of design procedure.**	**Adequate literature survey. Problem analysis performed correctly. Project demonstrates some use of design process.**	**Clear and complete literature survey, effective problem analyses is performed to choose design parameters. Project demonstrates effective use of design process.**			
Language, Grammar and References	**A lot of spelling and grammatical mistakes with poor English. The list of references is clearly inadequate. Table of content missing.**	**Frequent spellings and grammatical errors. The list of references should be expanded.**	**Occasional spellings and grammatical errors. The list of references appears reasonable but citation does not follow standard format.**	**Very few spellings and grammatical errors. Organization is good. The list of references appears reasonable and citation follow standard format.**	**Almost no spelling or grammatical mistake. Excellent organization. A comprehensive list of references is cited using the standard format.**			

Viva Voce

Assessment Criteria	Very Poor 0-1	Poor 2	Satisfactory 3	Good 4	Excellent 5	Scor e	Scor e	Scor e
						Student 1	Student 2	Student 3
Knowledge of Project Implementation details (Q/A)	No or very poor knowledge of implementation and design process.	Poor knowledge of implementation and design with wrong/no answers	Satisfactory knowledge of implementation, vague answers	Adequate knowledge of project implementation with majority of correct answers	Exceptional knowledge of implementation and overall design with clear and spontaneous answers.			